Annual report — Scientific research and collecting permit 26-635-017

Diversity of coralline algae in northern California and their reproductive systems
Principal investigator: Dr. Alena Reyes, Dept. of Biological Sciences, Cal Poly Humboldt
Permit term: Jul 22, 2026 – Jul 22, 2027 · Report period: full permit term

1. Summary

Field work ran from August 2026 through June 2027 across five state park units in the North Coast Redwoods and Mendocino districts. We completed 13 of 13 scheduled field visits: 10 intertidal surveys at low tides below –0.5 ft and 3 subtidal dive surveys at Van Damme SP. All collection stayed within the authorized quantities, and no rare, threatened, or endangered species were collected or disturbed.

Molecular identification (*psbA* and COI-5P barcodes) is complete for 98 of the 105 specimens. Early results confirm 12 coralline species at the study sites, including the first barcode-verified records of *Bossiella chiloensis* north of Point Arena.

2. Field visits

Park unit	Visits	Dates	Method
Tolowa Dunes SP	2	Aug 13–15, 2026; Apr 6, 2027	Intertidal transects
Del Norte Coast Redwoods SP	2	Sep 10, 2026; May 4, 2027	Intertidal transects
Prairie Creek Redwoods SP	2	Oct 8, 2026; May 22, 2027	Intertidal transects
MacKerricher SP	4	Nov 6, 2026 – Jun 2, 2027	Intertidal transects, reproductive sampling
Van Damme SP	3	Dec 3, 2026; Mar 18, 2027; Jun 14, 2027	SCUBA to 10 m, surface tender on shore

3. Specimens collected

Species	Authorized	Collected	Portion
Bossiella chiloensis	5	5	Single frond
Bossiella dichotoma	5	4	Single frond
Bossiella orbigniana	5	5	Single frond

Species	Authorized	Collected	Portion
Calliarthron tuberculosum	5	5	Single frond
Corallina chilensis	5	5	Single frond
Corallina vancouveriensis	50	47	Single frond and entire individual
Lithophyllum dispar	5	5	Single frond
Lithophyllum impressum	5	3	1 cm²
Lithothamnion phymatodium	5	5	1 cm²
Melobesia mediocris	5	5	Single frond
Mesophyllum conchatum	5	4	1 cm²
Neopolyporolithon reclinatum	5	5	Single crust
Total	**105**	**98**	

4. Compliance with special conditions

Before each visit we contacted the district permit coordinator, carried a copy of the permit in the field, and followed the plover protection measures at Tolowa Dunes SP: no work above the wrack line during nesting season and no vehicle access to the beach. Gear was cleaned and disinfected between sites. Divers kept the required distance from sea otters; one dive at Van Damme SP was cut short on Mar 18, 2027 when a raft of otters moved into the survey area.

5. Disposition of specimens

Voucher specimens are pressed and accessioned at the Cal Poly Humboldt Vascular Plant and Algal Herbarium (HSC-A 21407–21504). DNA extracts are stored at −80 °C in the Reyes lab, and sequences will be deposited in GenBank on publication.

6. Results to date

Corallina vancouveriensis showed reproductive conceptacles from February through May at MacKerricher SP, about six weeks later than at Tolowa Dunes SP. Two crustose specimens from Van Damme SP did not match published barcodes and may represent an undescribed *Lithophyllum* lineage; we are preparing morphological work to confirm.

7. Publications and presentations

Nadar, P., and A. Reyes. 2027. Barcoding northern California's articulated corallines. Western Society of Naturalists, poster. California Department of Parks and Recreation is acknowledged in the poster as required.

8. Plans

We plan to apply for a renewal to finish reproductive sampling of *C. vancouveriensis* through a second spring season, at the same sites and with no increase in collection quantities.

Submitted by Dr. Alena Reyes, Principal investigator
areyes@humboldt.edu · (707) 555-0142